

AWS CI/CD Pipeline for Order Cloud Lambda

1. Procedure

1.1 Create and maintain the source repository

Store the Lambda handler in the GitHub repository. The supplied repository screenshot shows `lambda_function.py` and `README.md`. The handler logs a deployment-verification message, reads `order_id` from the event (using `ORD-1001` as a fallback), logs the order being processed, and returns an HTTP-style `statusCode` of 200 with a success message and order ID.

1.2 Configure the CI/CD pipeline

Create the `OrderCloud-Lambda-CICD` pipeline with a GitHub source stage and an AWS Lambda deploy stage. The provided pipeline screenshot shows green success indicators for both stages.

1.3 Store and inspect source artifacts

Open the `SourceArt/` prefix in the pipeline's S3 bucket and confirm that source-artifact objects are present. The supplied screenshot shows three objects in this prefix.

1.4 Deploy and test the Lambda function

Open the `OrderCloudOrderProcessor` function in AWS Lambda. Run the saved test event named `OrderTest`. The supplied test output shows `Status: Succeeded`, `statusCode 200`, and the message "Order processed successfully."

1.5 Inspect runtime logs

Open the function's CloudWatch log stream. Verify that the log entries include the Lambda start, the message that the order processor started, processing of order `ORD-1001`, and the end/report records.

1.6 Verify Lambda versions

Inspect the Lambda Versions tab. The screenshot shows versions 1 through 4; version 1 has the `PRODUCTION` alias. This demonstrates that published versions are present, but it does not by itself prove that every version was invoked successfully.

1.7 Configure success-event notification

Use an EventBridge rule named `OrderCloud-Pipeline-Success`. Its event pattern matches source `aws.codepipeline`, detail-type `CodePipeline Pipeline Execution State Change`, state `SUCCEEDED`, and pipeline `OrderCloud-Lambda-CICD`. The rule is shown as Enabled.

1.8 Confirm email notification

Check the notification email. The provided AWS Notification Message reports the pipeline execution state as `SUCCEEDED` for `OrderCloud-Lambda-CICD`.

5. Screenshots and Observations

The following selected screenshots document the source repository, deployment, test response, artifact storage, runtime logs, Lambda versions, event rule, and notification. Repeated screenshots of the same Lambda Versions and CloudWatch pages were omitted to keep the report concise.

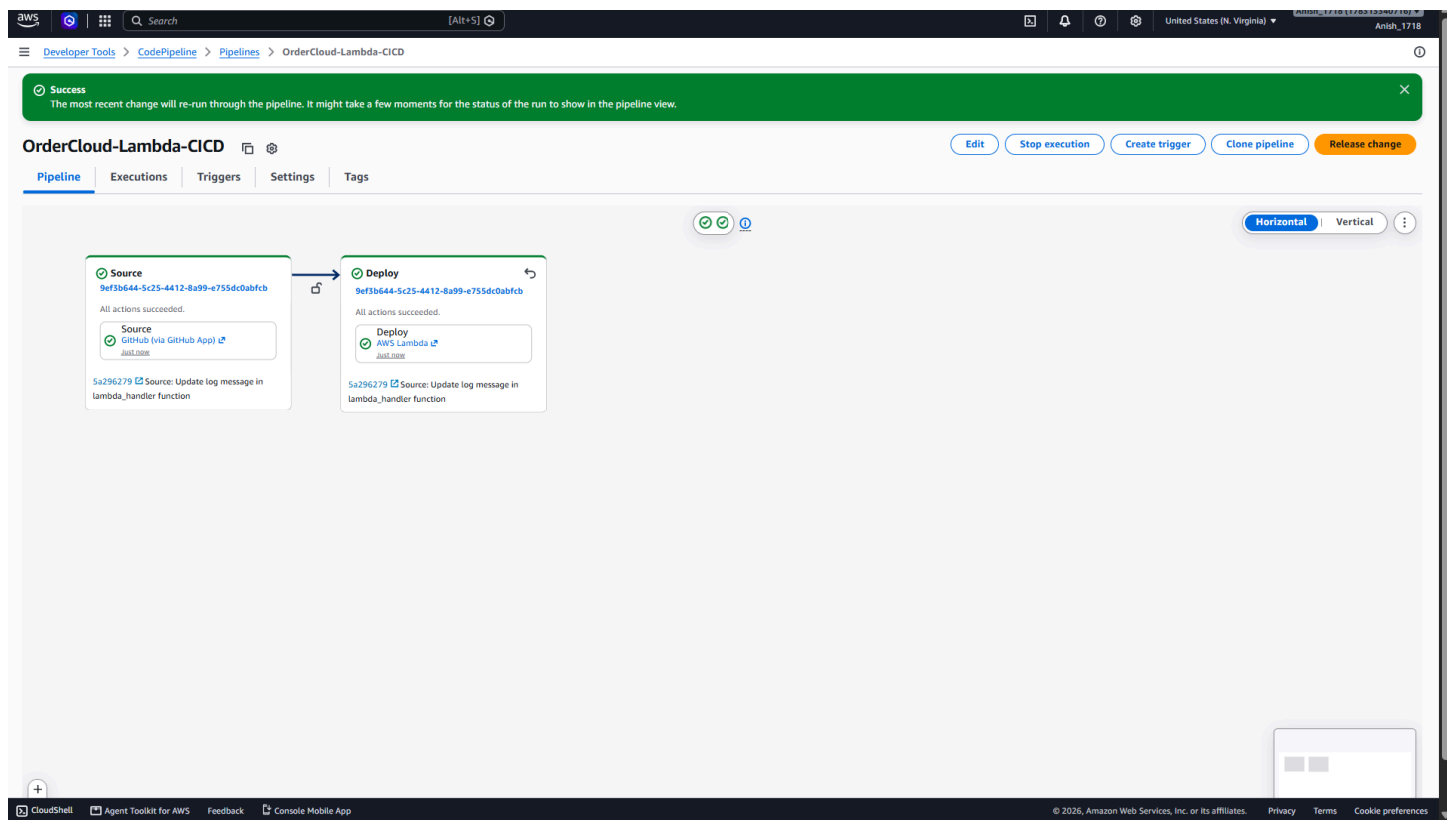


Figure 1. AWS CodePipeline showing successful Source and Deploy stages.

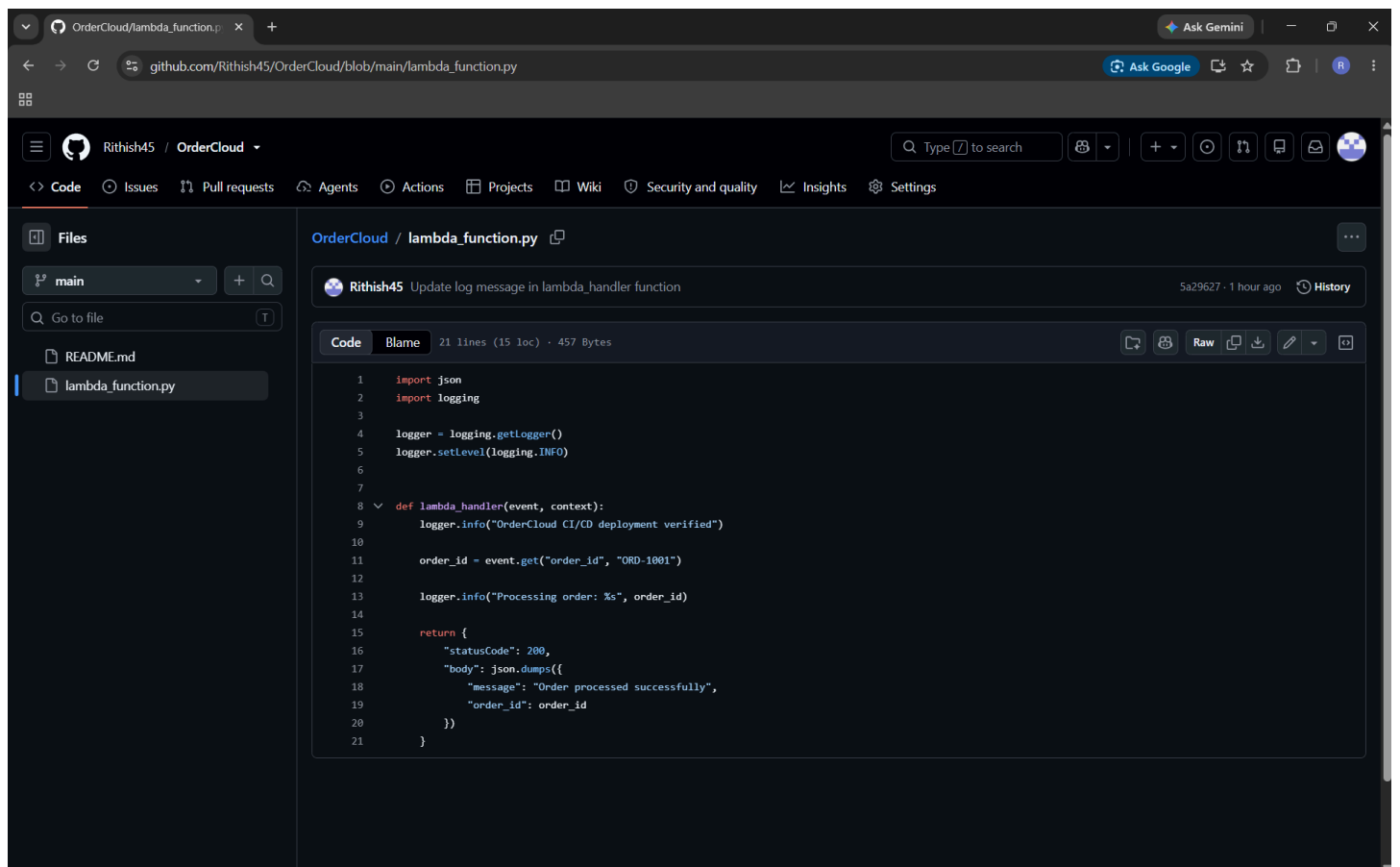


Figure 2. GitHub repository containing the Lambda function source code.

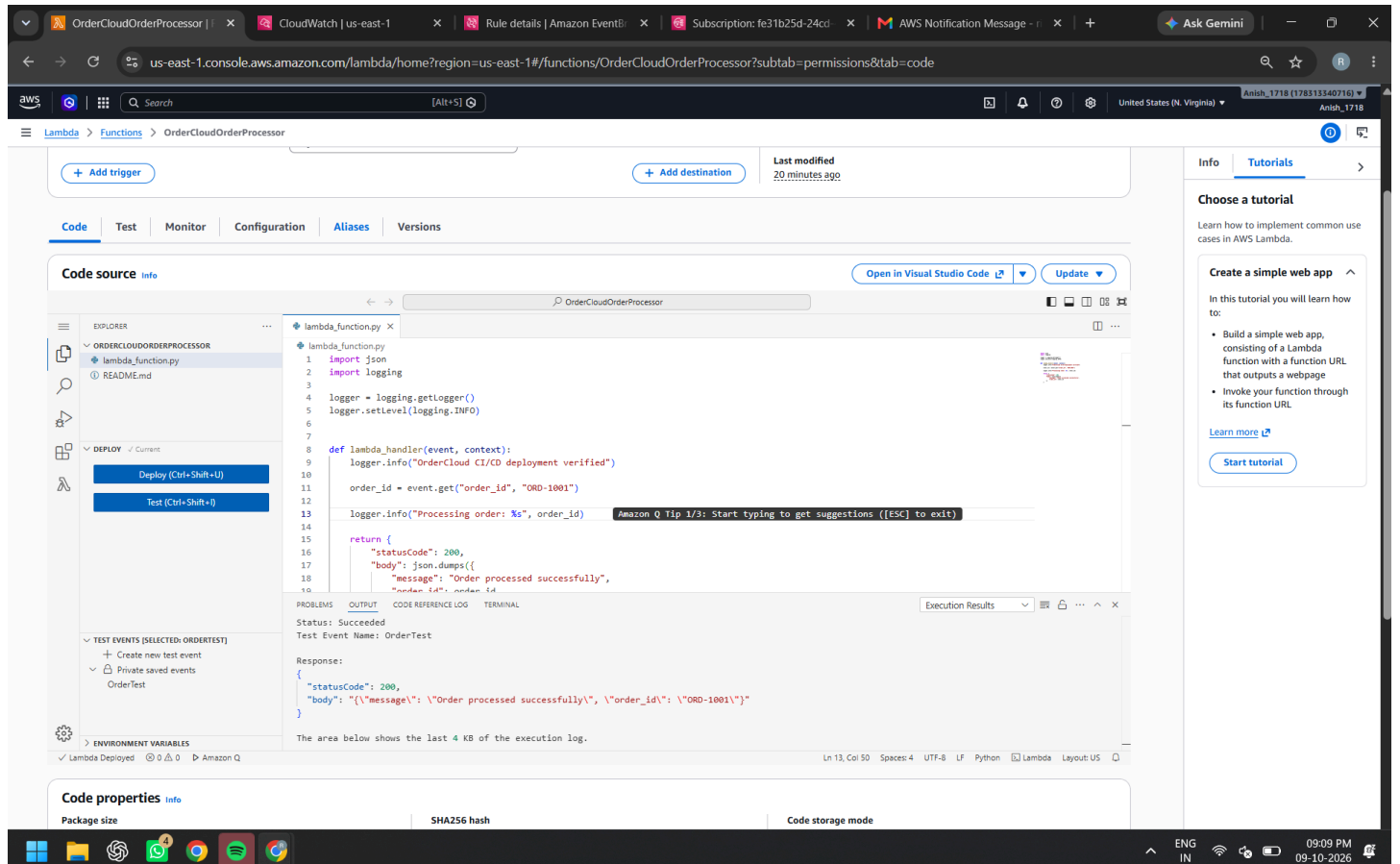


Figure 3. AWS Lambda code editor and successful test execution returning statusCode 200.

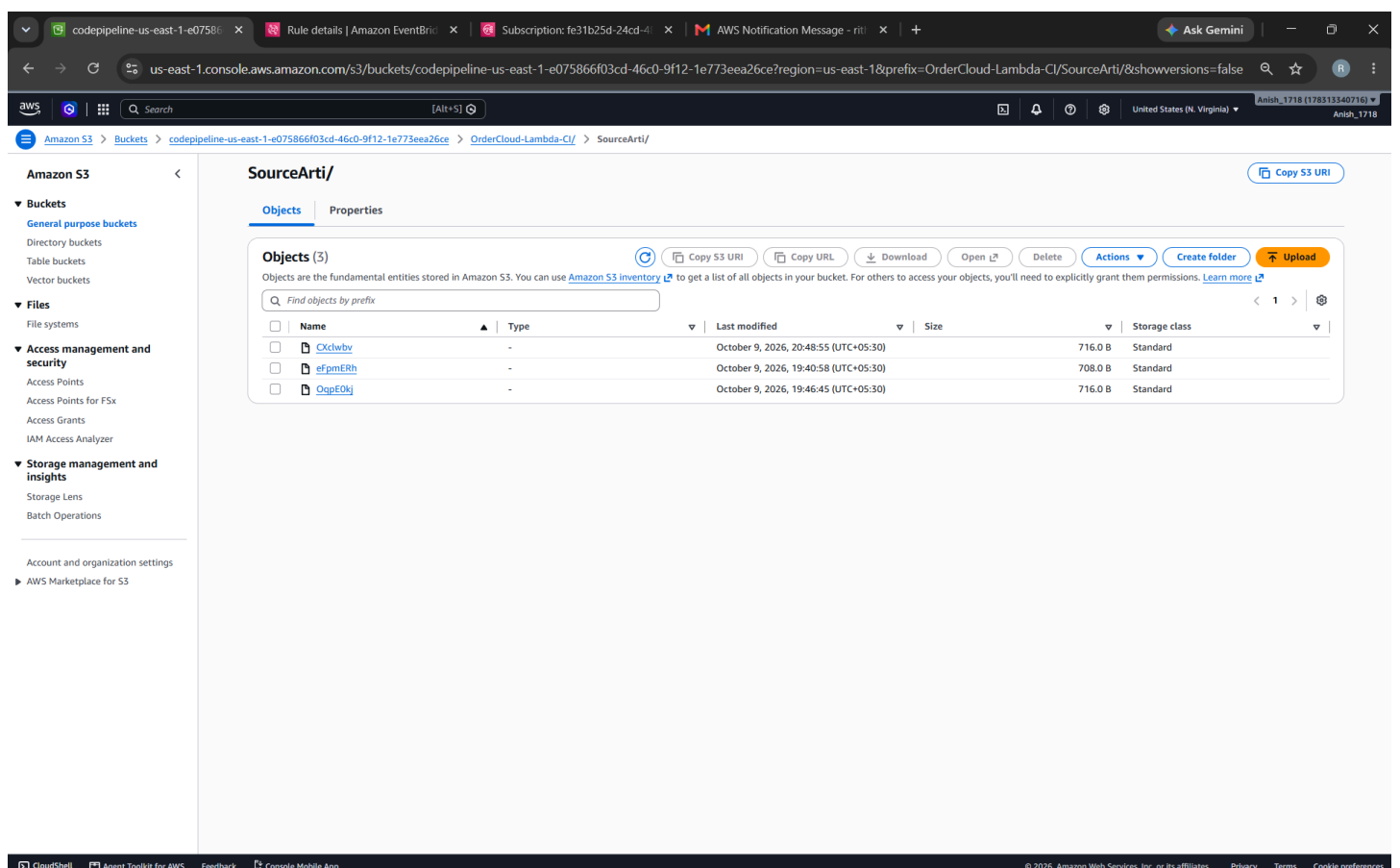


Figure 4. Amazon S3 SourceArt/ prefix containing the pipeline source artifacts.

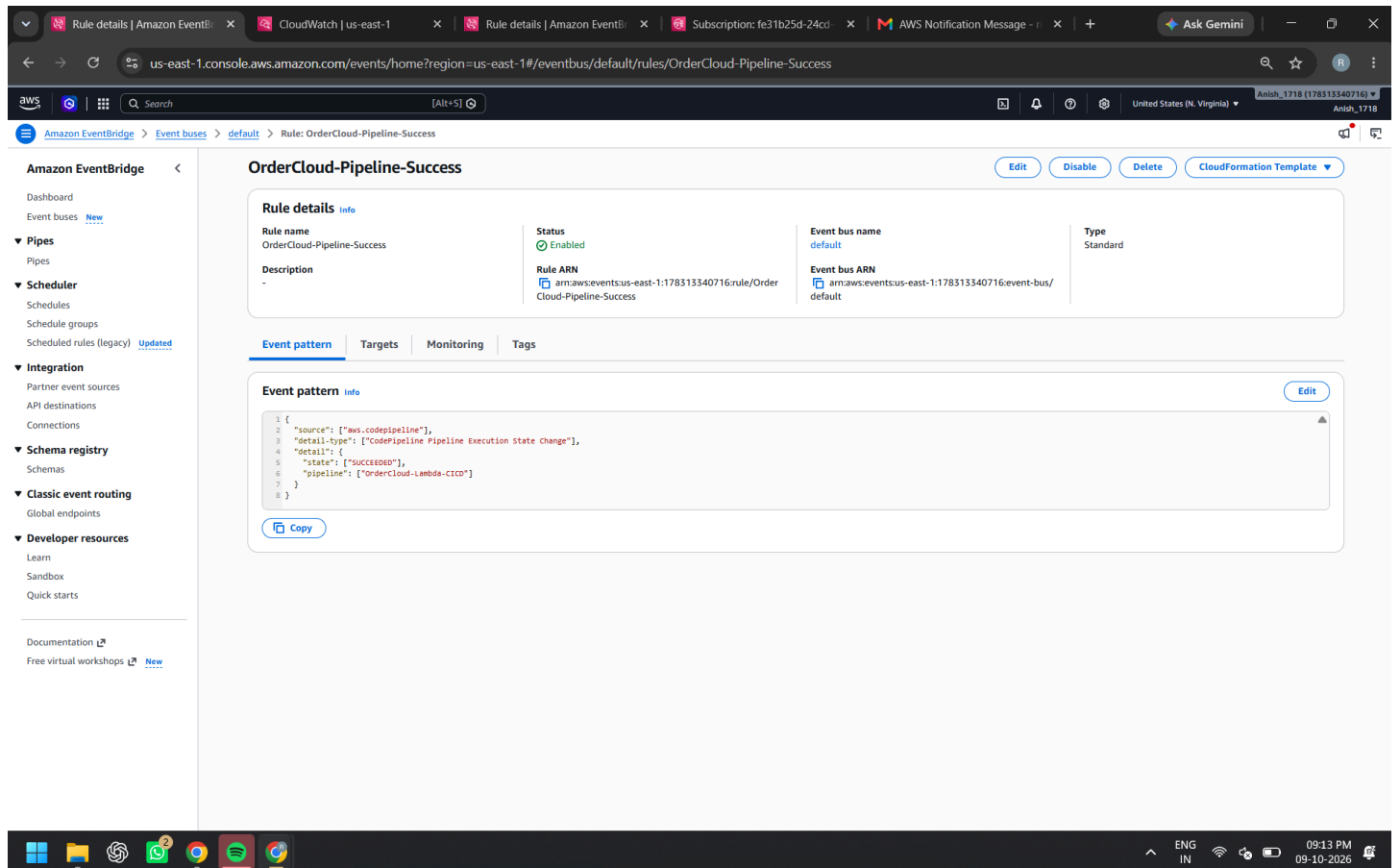


Figure 7. Amazon EventBridge rule configured for a successful CodePipeline execution.

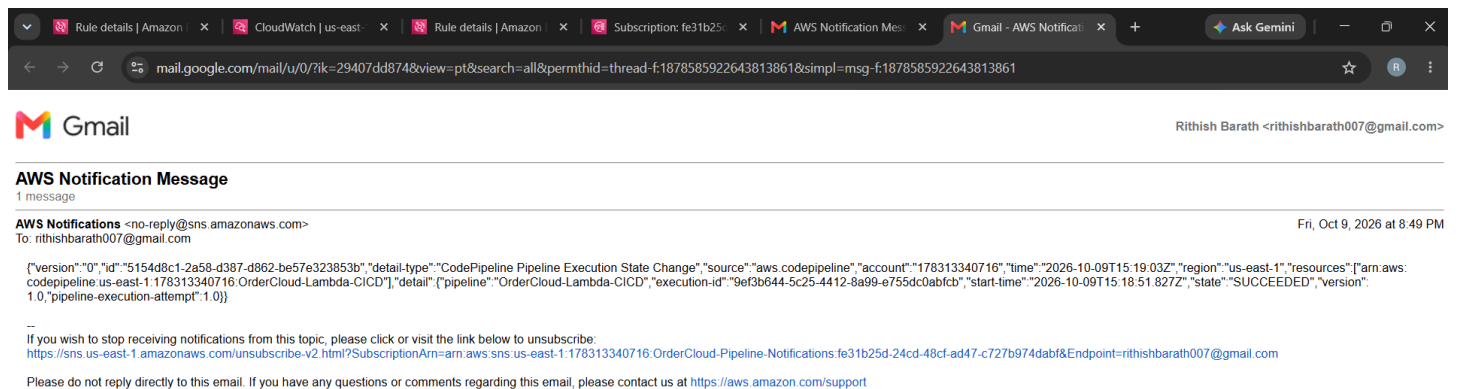


Figure 8. AWS SNS notification email reporting the pipeline execution state as SUCCEEDED.

